14:08:2026

Amlandeep's proposal:

$$p(\theta) > 0$$

$$1 = \int_{-\infty}^{\infty} \frac{1}{p(\theta)} d\theta \quad \{p(\theta)\}$$

Take all such poly.

Is it a conj. family of priors for Cauchy?

Q. Find a conj. family for any density $f(x|\theta)$.

Just take class of all densities.

- But it isn't useful, it will be useful when family is smaller.

Risk function: Stat T & parameter θ

$$R(\theta, T) = \mathbb{E}\{T-\theta\}^2$$

Baye's Risk: $\pi(\theta) \to$ Prior

$$r(\pi, T) = r_\pi(T) = \int_{\Theta} R(\theta, T)\pi(\theta) d\theta$$

$$= \mathbb{E}_\pi[R(\theta, T)]$$

Q. Check if Post. mean (Bayer's est.) minimizes Bayer's Risk.

Sol. $R(\theta, T) = \int_{\mathcal{X}} \{T(x) - \theta\}^2 f(x|\theta) dx$

$$r(\pi, T) = \int_{\Theta}\int_{\mathcal{X}} \{T(x)-\theta\}^2 f(x|\theta)\pi(\theta) dx\, d\theta$$

10 Aug 2026

12 (i) Distribution of quadratic forms.

$$= \int_{\mathcal{X}} \int_{\Theta} \{T(x) - \theta\}^2 \pi(\theta) f(x|\theta) \, d\theta \, dx$$

$$= \mathbb{E}\, \mathbb{E} \{ [T - \theta]^2 \mid X \}$$

$\mathbb{E}$ (outer): Marginal of X

$\mathbb{E}$ (inner): cond. expectation given X

= Minimised when inner expectation is minimised

i.e. when $T = \mathbb{E}[\theta|X]$

= Baye's estimate

Result: $X \in \mathcal{X}$ $\quad \theta \in \Theta$

$f(x|\theta)$ — Likelihood

$\pi(\theta)$ — Prior

$r(\pi, T)$ is minimized when we take

$T = \mathbb{E}(\theta|X)$ = Posterior mean

(Baye's Estimate)

- Example: We toss a coin

HHHTHHT

$\mathbb{P}(H) = p = \theta$

p: unknown parameter

$\hat{\theta} = \hat{p} = 5/7$

10 Aug 2026

(i) Distribution of quadratic forms.

"N.B. Problem" "Toss until 2 Tails"

$$f(x\mid\theta) = \binom{x-1}{1}\theta^{x-2}(1-\theta)^2$$

No Unbiased estimator exists for θ!

Example: $X \sim \mathcal{N}(\theta,1) = f(x\mid\theta)$

$\pi(\theta)=1 \;\forall\, \theta\in\mathbb{R}$. Prior might not be a proper density.
(Everything is equally likely)

Sol. $\rightarrow$ Improper prior (don't integrate to 1)

Not a proper density

$\pi(\theta\mid X) = \mathcal{N}(X,1) \Rightarrow$ Generalized Baye's Estimate $= X$

A proper density

Q Suppose we have n iid obs.. What is Gen Baye's est. w.r.t. improper prior $\pi(\theta)=1$?

$X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta,1)$

$\pi(\theta)=1 \;\forall\,\theta\in\mathbb{R}$

what is a gen. Baye's est.?

Sol. $\overline{X}$

- Gen. Baye's estimate are functions of suff. stat. (by NFFT).

Example: $X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} \dfrac{1}{\pi\{1+(x-\theta)^2\}}$

what is ML eq?

We get a poly eq. with no closed sol..

5)

10 Aug 2026

(i) Distribution of quadratic forms.

$U \sim \text{Unif}\,[0, 1]$

$X = \tan\{\pi\,(U - \tfrac{1}{2})\} + \theta$

Qn: $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \dfrac{1}{\pi\{1+(x-\theta)^2\}}$

Is there any unbiased estimate for θ?

- For $n = 1$, certainly not!

ob.

numbers

S

$= 0$